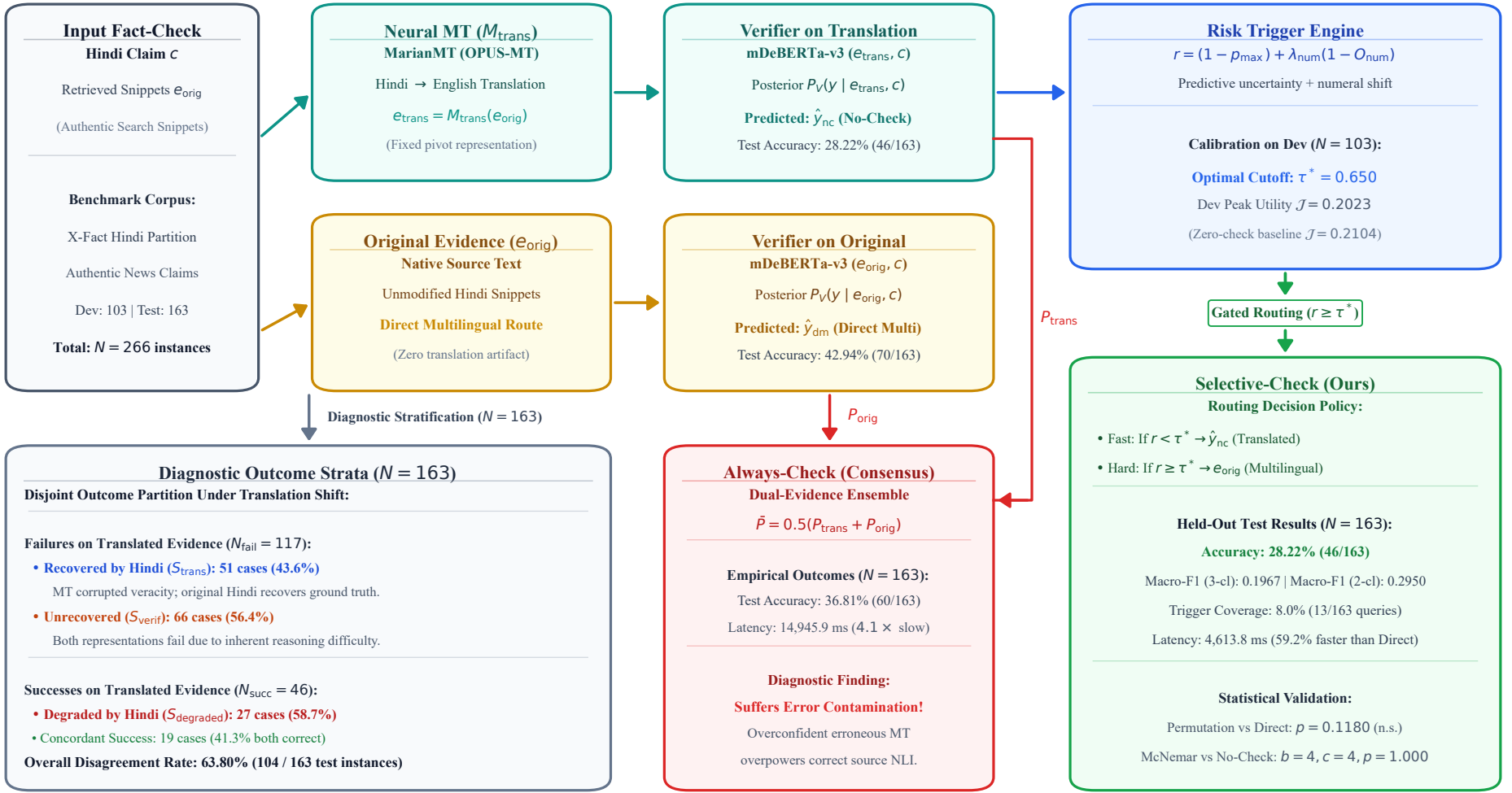

Input Fact-Check
Hindi Claim c
Retrieved Snippets e_{orig}
(Authentic Search Snippets)
Benchmark Corpus:
X-Fact Hindi Partition
Authentic News Claims
Dev: 103 | Test: 163
Total: $N = 266$ instances
Neural MT (M_{trans})
MarianMT (OPUS-MT)
Hindi → English Translation
$e_{trans} = M_{trans}(e_{orig})$
(Fixed pivot representation)
Original Evidence (e_{orig})
Native Source Text
Unmodified Hindi Snippets
Direct Multilingual Route
(Zero translation artifact)
Verifier on Translation
mDeBERTa-v3 (e_{trans}, c)
Posterior $P_V(y \mid e_{trans}, c)$
Predicted: $\hat{y}_{nc}$ (No-Check)
Test Accuracy: 28.22% (46/163)
Verifier on Original
mDeBERTa-v3 (e_{orig}, c)
Posterior $P_V(y \mid e_{orig}, c)$
Predicted: $\hat{y}_{dm}$ (Direct Multi)
Test Accuracy: 42.94% (70/163)
Risk Trigger Engine
$r = (1 - p_{max}) + \lambda_{num}(1 - O_{num})$
Predictive uncertainty + numeral shift
Calibration on Dev ($N = 103$):
Optimal Cutoff: $\tau^* = 0.650$
Dev Peak Utility $\mathcal{J} = 0.2023$
(Zero-check baseline $\mathcal{J} = 0.2104$)
P_{trans}
Gated Routing ($r \geq \tau^*$)
P_{orig}
Diagnostic Stratification ($N = 163$)
Diagnostic Outcome Strata ($N = 163$)
Disjoint Outcome Partition Under Translation Shift:
Failures on Translated Evidence ($N_{fail} = 117$):
• Recovered by Hindi (S_{trans}): 51 cases (43.6%)
MT corrupted veracity; original Hindi recovers ground truth.
• Unrecovered (S_{verif}): 66 cases (56.4%)
Both representations fail due to inherent reasoning difficulty.
Successes on Translated Evidence ($N_{succ} = 46$):
• Degraded by Hindi ($S_{degraded}$): 27 cases (58.7%)
• Concordant Success: 19 cases (41.3% both correct)
Overall Disagreement Rate: 63.80% (104 / 163 test instances)
Always-Check (Consensus)
Dual-Evidence Ensemble
$\bar{P} = 0.5(P_{trans} + P_{orig})$
Empirical Outcomes ($N = 163$):
Test Accuracy: 36.81% (60/163)
Latency: 14,945.9 ms (4.1 × slow)
Diagnostic Finding:
Suffers Error Contamination!
Overconfident erroneous MT
overpowers correct source NLI.
Selective-Check (Ours)
Routing Decision Policy:
• Fast: If $r < \tau^* \rightarrow \hat{y}_{nc}$ (Translated)
• Hard: If $r \geq \tau^* \rightarrow e_{orig}$ (Multilingual)
Held-Out Test Results ($N = 163$):
Accuracy: 28.22% (46/163)
Macro-F1 (3-cl): 0.1967 | Macro-F1 (2-cl): 0.2950
Trigger Coverage: 8.0% (13/163 queries)
Latency: 4,613.8 ms (59.2% faster than Direct)
Statistical Validation:
Permutation vs Direct: $p = 0.1180$ (n.s.)
McNemar vs No-Check: $b = 4, c = 4, p = 1.000$